Network Routing PBQ - Answer Key

Explained key for the attached Network+ routing PBQ

Core idea: R1 chooses the most specific matching route in its table. If no specific route matches, it uses the default route.

Quick topology reference

Device / Route	Key network	Reach via	Why it matters on R1
LAN A	192.168.10.0/24	Connected G0/0	This is local to R1, so R1 does not send this traffic to another router.
LAN B behind R2	172.16.20.0/24	Next hop 10.0.0.2	R1 has a specific static route pointing to R2 for this network.
LAN C behind R3	192.168.30.0/24	Next hop 10.0.0.6	R1 has a specific static route pointing to R3 for this network.
Internet / unknown destinations	0.0.0.0/0	Next hop 10.0.0.6	Anything without a more specific match uses the default route through R3 toward the ISP.

1. A host on LAN A (192.168.10.0/24) needs to reach a server on LAN B (172.16.20.0/24). Which next hop or interface should R1 use?

Correct answer: **B) Next hop 10.0.0.2 (R2)**

Why it is correct: R1 has an explicit static route for 172.16.20.0/24 with next hop 10.0.0.2. Because that route directly matches LAN B, R1 forwards the packet to R2, and R2 then delivers it onto LAN B.

Routing logic: Best-match rule: 172.16.20.0/24 is more specific than 0.0.0.0/0.

Why the other choices are wrong:

- A) Send directly out G0/0 - G0/0 is R1's local interface for LAN A, not the path to LAN B.
- C) Next hop 10.0.0.6 (R3) - R3 is the path to LAN C and the default route, but there is a more specific route for LAN B via R2.
- D) Use the default route to 10.0.0.6 - Routers prefer a specific matching route over the default route, so the default is not used here.

2. A user on LAN A (192.168.10.0/24) needs to access a file share on LAN C (192.168.30.0/24). Which next hop or interface should R1 use?

Correct answer: **B) Next hop 10.0.0.6 (R3)**

Why it is correct: R1 has a static route for 192.168.30.0/24 via 10.0.0.6. That means any traffic destined for LAN C should be forwarded to R3.

Routing logic: Specific route beats default route, even when both point to the same next hop.

Why the other choices are wrong:

- A) Next hop 10.0.0.2 (R2) - R2 is only the correct next hop for LAN B, not for LAN C.
- C) Send directly out G0/0 - G0/0 is the connected interface for LAN A, so using it would keep the traffic on the wrong network.
- D) Use the default route to 10.0.0.6 - Although the next hop is the same as the default route, the correct reason is the specific static route to 192.168.30.0/24. On an exam, choose the most specific correct route, not just a route that happens to work.

3. A workstation on LAN A (192.168.10.0/24) is browsing an external website on the Internet. Which route should R1 use to forward this traffic?

Correct answer: **C) Default route 0.0.0.0/0 via 10.0.0.6**

Why it is correct: The Internet destination is not one of R1's directly connected networks and does not match the specific static routes to LAN B or LAN C. Because no more specific route exists, R1 uses its default route to 10.0.0.6, sending the traffic to R3, which has the path to the ISP.

Routing logic: Use the default route when no connected or more specific static route matches the destination.

Why the other choices are wrong:

- A) Static route to 172.16.20.0/24 via 10.0.0.2 - That route is only for LAN B.
- B) Static route to 192.168.30.0/24 via 10.0.0.6 - That route is only for LAN C, not for all outside destinations.
- D) Connected route 192.168.10.0/24 via G0/0 - That connected route is for R1's local LAN, not for traffic leaving toward the Internet.

Exam reminder

- Connected routes are used for networks directly attached to the router.
- Specific static routes are preferred over the default route when both could match.
- The default route is the fallback path for destinations not otherwise listed in the routing table.